MAYBEWELL BOOKS

Space STEM Pack

Twelve hands-on projects for future astronomers and engineers

A CREATIVITY BOOK

no kit required.

maybewellbooks.com

HOW TO USE THIS BOOK

Mission Briefing

Every project here does real science with paper, a pencil, and sometimes scissors and sunlight.

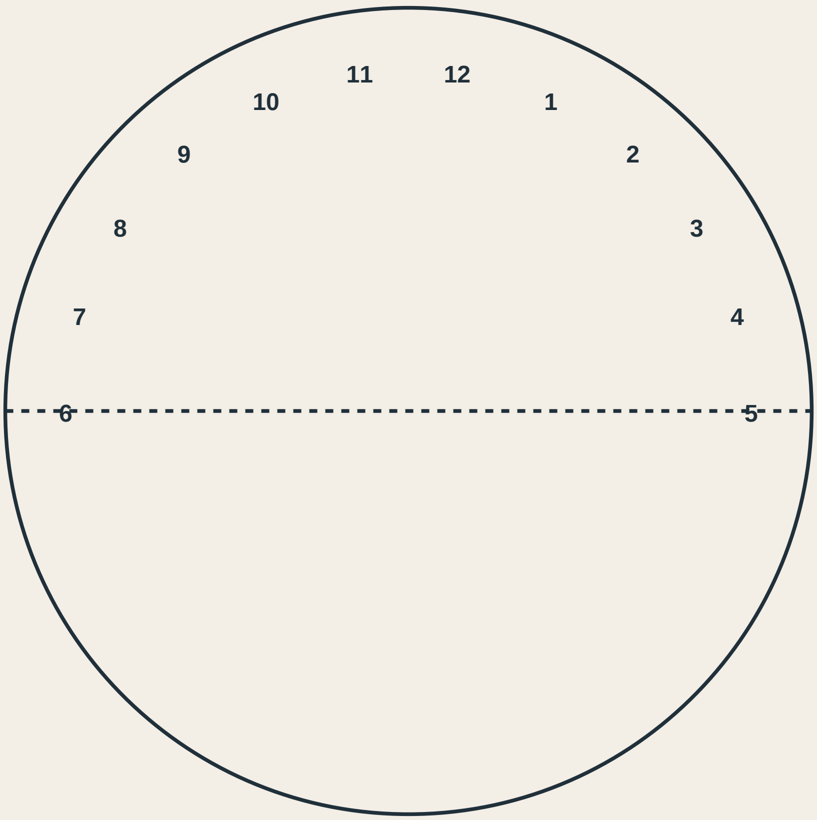
Some pages are math. Some are charts you fill in over real nights outside. Some are things you build. None of them need a kit.

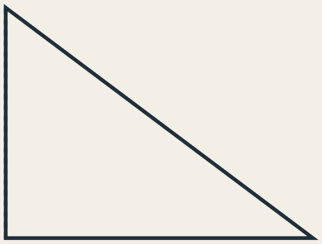
Teacher and parent notes are in the back — what each project teaches, and how long it usually takes.

PROJECT #1 OF 12

Build a Paper Sundial

Cut out the gnomon (the triangle) and fold it upright along the dashed line on the dial. Set it outside facing true north, and check it every hour on a sunny day.





cut solid, fold dashed (gnomon)

PROJECT #2 OF 12

Chart the Big Dipper

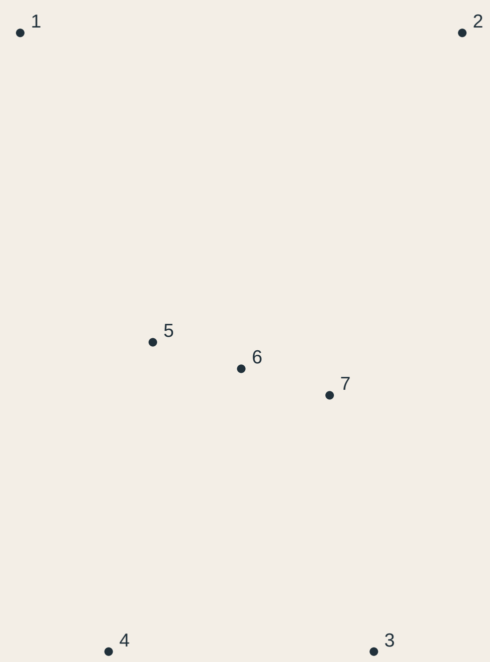
Connect the stars in order to trace the Big Dipper, part of the constellation Ursa Major. Look for it in the northern sky after dark.

The Big Dipper

PROJECT #3 OF 12

Chart Orion

Orion the Hunter is one of the easiest constellations to find — look for three stars in a short, straight line. That's his belt.

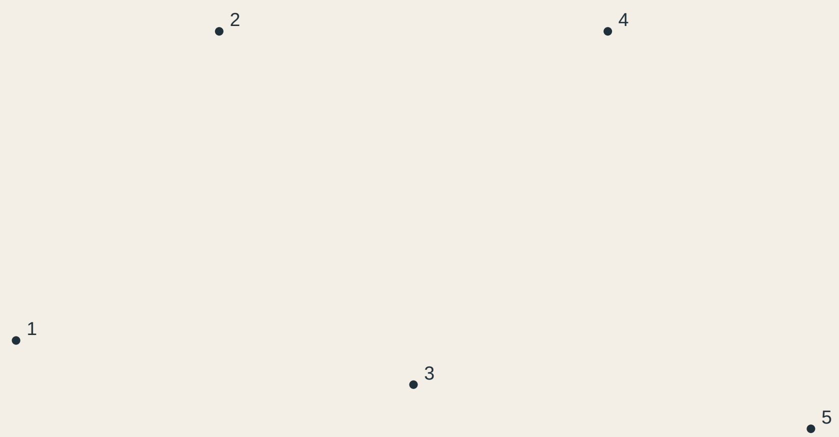


Orion

PROJECT #4 OF 12

Chart Cassiopeia

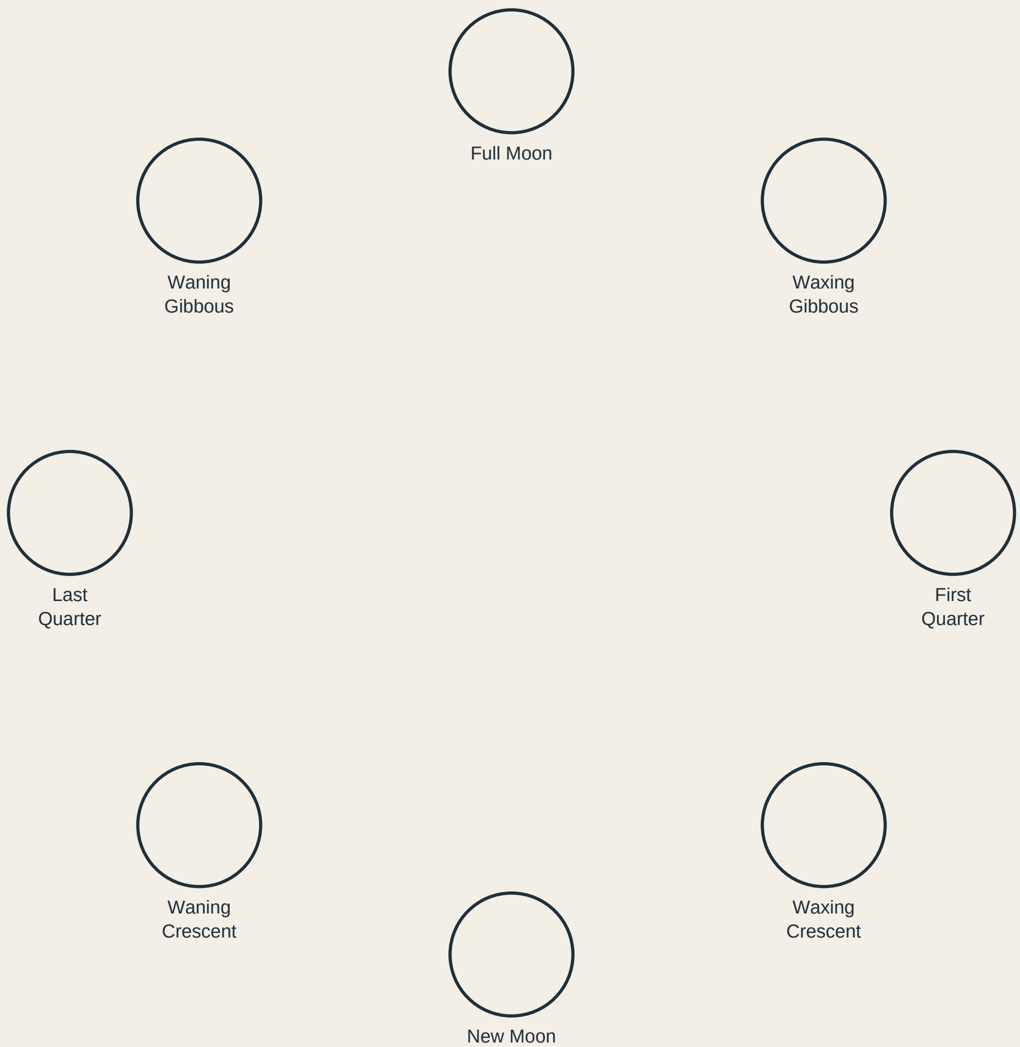
Cassiopeia looks like a wide letter W (or M, depending which way is up). It's visible year-round from most of the northern hemisphere.



Cassiopeia

Moon Phase Wheel

Color each moon shape to match its phase name, then cut out the wheel and the window card, and pin them together in the center so the wheel spins.



PROJECT #6 OF 12

Rocket Math: Fuel Check

Real rockets are mostly fuel by weight. Work through the problems to see just how much.

1. A rocket weighs 2,970,000 kg fully fueled. Empty, it weighs 130,000 kg. How many kg of fuel is that?

Answer: ______________

2. If fuel burns at 8,400 kg per second, how many seconds until the tank above is empty?

Answer: ______________

3. A smaller rocket carries 45,000 kg of fuel and uses 15,000 kg per stage. How many stages can it fuel?

Answer: ______________

4. Mission control needs a 12% fuel reserve on a 60,000 kg fuel load. How many kg is the reserve?

Answer: ______________

PROJECT #7 OF 12

Rocket Math: Countdown

Practice the kind of quick arithmetic mission control does under pressure.

1. T-minus 45 seconds. A system check takes 12 seconds. How many seconds remain after the check?

Answer: ______________

2. Three checks of 8 seconds each must finish before T-minus 10. What's the latest they can start?

Answer: ______________

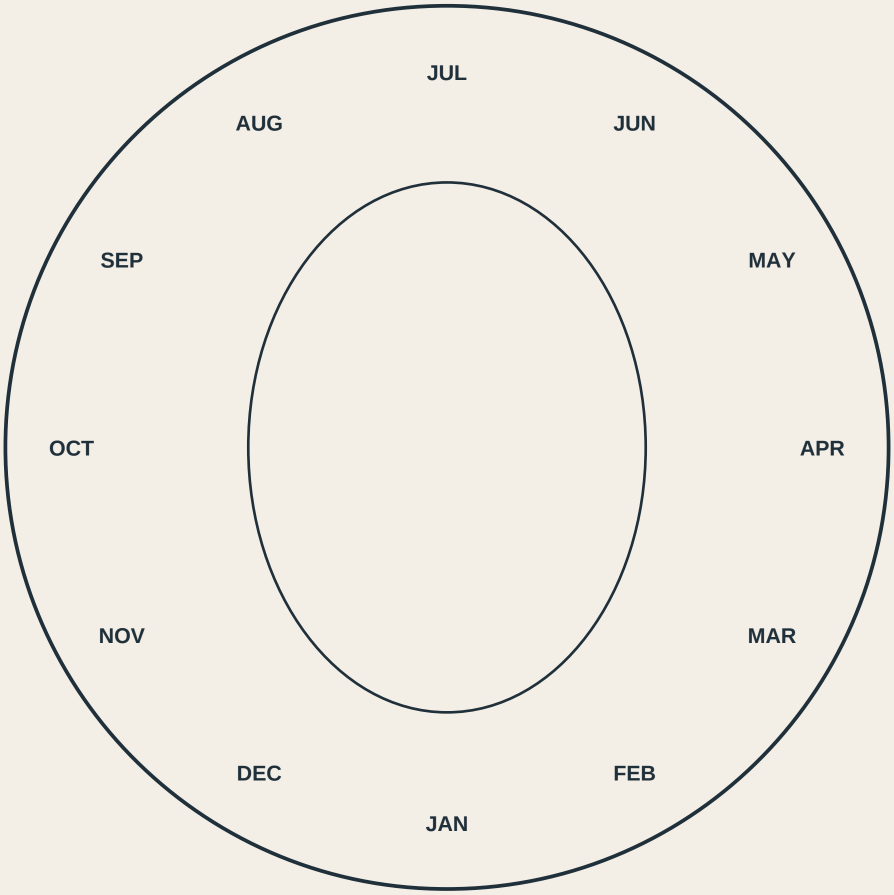
3. The countdown holds for 90 seconds at T-minus 20. What is the new T-minus after the hold ends and 90 more seconds pass?

Answer: ______________

PROJECT #8 OF 12

Build a Star Finder

Cut out the wheel and the cover, layer them, and rotate to the current month and time — the oval window shows which constellations are up tonight.



cut out both circles, layer, pin at the center, and rotate

PROJECT #9 OF 12

Scale of the Solar System

If the Sun were the size shown here, how far away would each planet really be? Fill in your guesses before checking the answers at the bottom.

Sun	the size of a beach ball
Mercury	a peppercorn, 12 steps away
Venus	a pea, 22 steps away
Earth	a pea, 30 steps away
Mars	a peppercorn, 46 steps away
Jupiter	a grapefruit, 155 steps away
Saturn	an orange, 287 steps away

PROJECT #10 OF 12

Design Your Own Spacecraft

Every spacecraft needs to solve the same problems: power, life support, propulsion, and coming home. Label your design.

Label these on your design: Power source • Life support • Propulsion • Landing / return

PROJECT #11 OF 12

Gravity Drop Experiment

Drop two different objects from the same height at the same time. Record what you predicted and what actually happened.

Object	My prediction	What happened
Object 1		
Object 2		
Object 3		

PROJECT #12 OF 12

Name Your Own Constellation

Connect these unnamed stars into a shape, give it a name, and write the myth behind it.

Constellation name: ______________________

FOR PARENTS & TEACHERS

NOTES

TEACHER NOTES

Projects 1, 5, 8

Best done a day ahead — sundial and star finder need daylight or a clear night to test.

Projects 2-4, 12

No equipment needed. Great for a car ride or a waiting room.

Projects 6-7, 9

Calculator optional but not required — numbers are chosen to stay simple.

Project 10

No wrong answers. Encourage kids to explain their choices out loud.

Project 11

Needs two small objects of different weight and a safe place to drop them.